

Sid Agrawal

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EXPERIENCE

Staff Software Engineer - Operating Systems Virtualization VMware by Broadcom

- 📅 07/2025 - Present [📍 Bellevue, WA, USA](#)
- Developed VMware ESXi hypervisor with a focus on confidential computing (TDX), and virtual machine partitioning (Microsoft VSM).
 - Developed paravisor used inside a partitioned confidential VM.
 - Extended ESXi to support VSM hypercalls for kernel partitioning.

Doctoral Researcher - Operating Systems Architecture Systopia Lab, University of British Columbia

- 📅 01/2021 - Present [📍 Vancouver, BC, Canada](#)
- Bootstrapped a research project, developing a prototype OS to compare various isolation mechanisms like Docker, Kata, VM (Xen, KVM) from an isolation and security standpoint, extending to new discoveries.
 - Led a team of three engineers to develop an OS using 50K SLOC in C and ARM assembly on the seL4 microkernel, integrating hypervisor, device drivers, and isolation mechanisms with complete [Source Code and Documentation](#).
 - Developed Python tooling to analyze isolation mechanisms on Linux through Namespaces, Docker, QEMU, and Buildroot, with an accompanying [Code and Wiki](#).
 - Analyzed large-scale graphs in Neo4j to present isolation mechanism differences using CypherQL.

Intern, Security Research - Operating Systems Security ARM

- 📅 05/2022 - 08/2022 [📍 Remote, Canada](#)
- Ported the seL4 microkernel to ARM's Morello experimental platform with hardware capability support (CHERI). [Blog](#).
 - Investigated kernel capability system, bootloader, context switching, and process bootstrapping code paths.

Software Engineer - Microservices and Tooling Arista Networks

- 📅 09/2016 - 12/2020 [📍 SF Bay Area, USA & Vancouver, BC, Canada](#)
- Developed (Golang) and deployed (Kubernetes and Jenkins) microservices to detect, report (Grafana), triage, and fix faulty test beds. This automation led to savings of 10s of person-hours per month per engineer. Expanded it from a solo project to a 3 member team
 - Developed (Golang) and deployed services to store distributed file system's block data in a NoSQL (ScyllaDB) store
 - Participated in DevOps responsibilities, for the Kubernetes, Redis, Swift and ScyllaDB clusters

Software Engineer - Distributed File Systems Panzura

- 📅 04/2015 - 08/2016 [📍 SF Bay Area, USA](#)
- Designed and implemented (C) support to transactionally update file metadata for Panzura Global Distributed File System (ZFS on FreeBSD). This minimized recovery after crashes, thus preventing an entire class of support tickets.

Software Engineer - Solaris OS Oracle/Sun

- 📅 03/2012 - 04/2015 [📍 SF Bay Area, USA](#)
- Enhanced the virtual memory predictor in Solaris by developing an algorithm to determine which segments in the address space can be upgraded to large pages

EDUCATION

PhD in Computer Science The University of British Columbia

📅 01/2021 - Present [📍 Vancouver, BC, Canada](#)

MS in Electrical and Computer Engineering University of Florida

📅 08/2010 - 12/2011 [📍 Gainesville, Florida, USA](#)

BE in Electrical and Electronics Engineering Birla Institute of Technology and Science, Pilani

📅 08/2005 - 08/2009 [📍 Goa, India](#)

PUBLICATIONS AND TALKS

Exploring the Design Space of Isolation Mechanisms

PLOS 2026, Prague, Czechia

Sidhartha Agrawal, Shaurya Patel, Linh Pham, Arya Stevinson, Ilias Karimalis, Hugo Lefeuvre, Aastha Mehta, Reto Achermann, Margo Seltzer

<https://doi.org/10.1145/3831586.3838145>

Comparing Isolation Mechanisms with OSmosis

PLOS 2025, South Korea

Sidhartha Agrawal, Shaurya Patel, Linh Pham, Arya Stevinson, Ilias Karimalis, Hugo Lefeuvre, Aastha Mehta, Reto Achermann, Margo Seltzer

<https://dl.acm.org/doi/10.1145/3764860.3768325>

CellulOS: An OS for comparing Isolation Mechanisms

seL4 Summit 2025, Prague, Czech Republic

Sid Agrawal, Arya Stevinson, Linh Pham, Ethan Xu, Shaurya Patel, Hugo Lefeuvre, Aastha Mehta, Reto Achermann, Margo Seltzer

<https://sel4summit2025.sched.com/event/26GFQ/>

Securing Monolithic Kernels using Compartmentalization

ArXiv 2024

Soo Yee Lim, Sidhartha Agrawal, Xueyuan Han, David Eysers, Dan O'Keefe, Thomas Pasquier

<https://arxiv.org/pdf/2404.08716>

CHERI-picking: Leveraging capability hardware for prefetching

PLOS 2022, Germany

Shaurya Patel, Sidhartha Agrawal, Alexandra Fedorova, and Margo Seltzer

<https://dl.acm.org/doi/pdf/10.1145/3623759.3624553>